

Summer 2026 Grassland Soil Core Figures

Created: August 16, 2026 Last Updated:

Jared Umphress

Table of contents

Part 1: Set up tasks	1
Part 2: Convert EC measurements to dS/m	2
2.1: Get mean EC, soil moisture, and sand/clay content values for depth midpoints .	3
Part 3: Plot EC and soil moisture based on transect	3
Part 4: Plot Average Sand% and Average Clay % by Depth	5

Part 1: Set up tasks

Load in libraries

```
library(tidyverse)
library(janitor)
library(here)

#plotting
library(cowplot)
library(ggthemes) # to use Paul Tol color palettes

# packages specifically for soil science
library(aqp) # algorithms for quantitative pedology
# library(soiltexture)
```

Load in data

```
soil_data <- read.csv(here("data/soil_samples/2026_summer_grassland_soil_cores.csv"))
```

Part 2: Convert EC measurements to dS/m

This function also accounts for dilution

```
source(here("code/mS_per_cm_to_dS_per_m.R"))

#clean data, get depth midpoints, and estimate EC in dS/m
soil_data <- soil_data %>%
  clean_names() %>%
  mutate(
    depth_midpoint = -(depth_top + depth_bottom)/2,
    ec_ds_per_m = uS_per_cm_to_dS_per_m(ec_u_s_cm),
    ec_converted = (ec_u_s_cm/1000)*5,
    year = year(sample_date),
    # use lapply to apply the conversion function to the vector
    ssc = lapply(texture_code, texcl_to_ssc),
    #subset ssc to the corresponding columns
    sand = sapply(ssc, `[`, "sand"),
    silt = sapply(ssc, `[`, "silt"),
    clay = sapply(ssc, `[`, "clay"),
    year = year(sample_date)
  ) %>%
  select(-ssc) %>% # Remove temporary column
  #move quantitative columns next to texture
  relocate(sand:clay, .after = texture_code)
```

Test uS/cm to dS/cm conversion

```
# high value of 2440
uS_per_cm_to_dS_per_m(2440)
```

```
[1] 16.451
```

```
# 16.451 dS/cm
```

```
# low value of 235 uS/cm
uS_per_cm_to_dS_per_m(235)
```

```
[1] 3.0005
```

```
# 3.005 dS/cm
```

2.1: Get mean EC, soil moisture, and sand/clay content values for depth midpoints

```
#calculate mean values for both plots combined
means_2026_summer <- soil_data %>%
  group_by(depth_midpoint) %>%
  summarize(mean_ec = mean(ec_ds_per_m, na.rm = TRUE),
            mean_moisture = mean(percent_soil_moisture),
            mean_sand = mean(sand),
            mean_clay = mean(clay),
            n = n()) %>%
  ungroup()
```

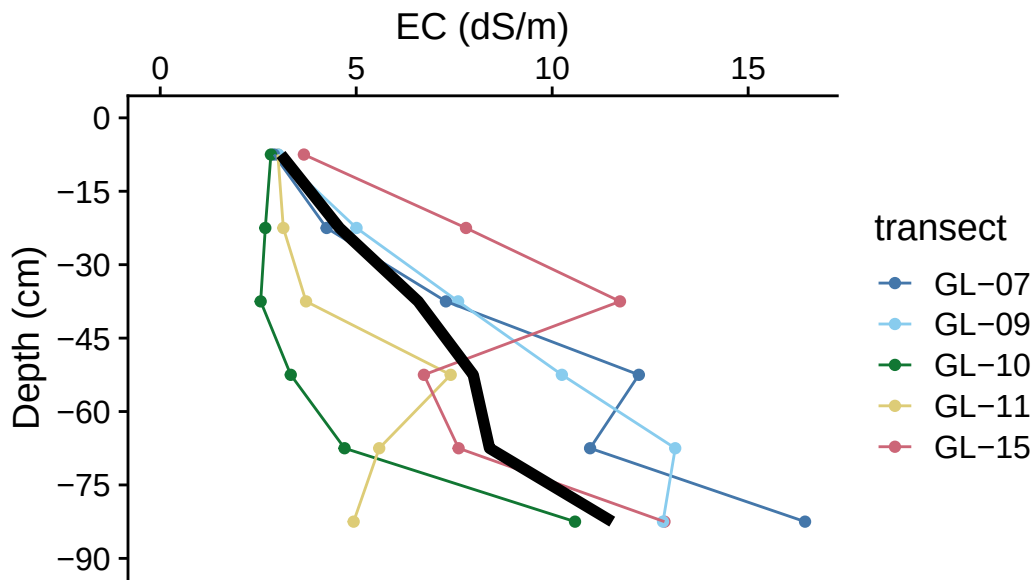
Part 3: Plot EC and soil moisture based on transect

EC figures

```
fig_2026_summer_ec <- ggplot(data = soil_data, aes(x = depth_midpoint, y = ec_ds_per_m, color = depth_midpoint)) +
  geom_point() +
  geom_path() +
  geom_path(data = means_2026_summer, aes(x = depth_midpoint, y = mean_ec), color = "black", size = 2) +
  coord_flip() +
  theme_cowplot() +
  scale_x_continuous(limits = c(-90,0), breaks = seq(-90, 0, by = 15)) +
  scale_y_continuous(limits = c(0,NA), position = "right") +
  labs(y = "EC (dS/m)", x = "Depth (cm)", title = "Mesa Grassland Cores, Summer 2026") +
  scale_color_ptol()

fig_2026_summer_ec
```

Mesa Grassland Cores, Summer 2026

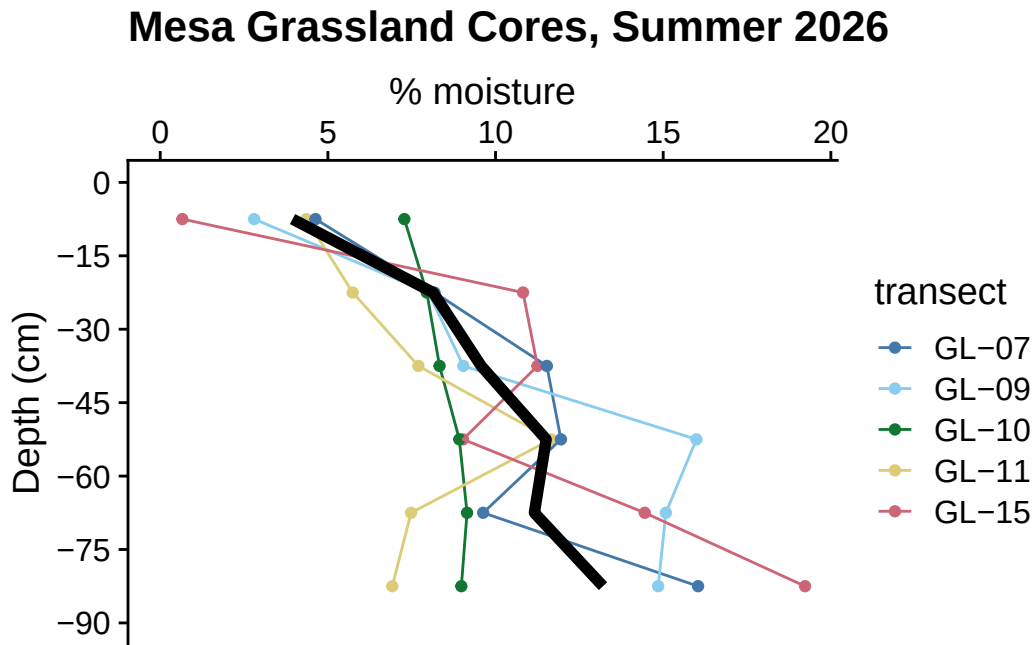


```
#save to file
ggsave(filename = here("figures/soil_cores/Summer_2026_Mesa_Grassland_EC.pdf"),
  fig_2026_summer_ec,
  width = 6,
  height = 4,
  units = "in",
  bg = "white")
```

Soil Moisture figures

```
# moisture plot ----
fig_2026_summer_moisture <- ggplot(data = soil_data, aes(x = depth_midpoint, y = percent_soil_moisture)) +
  geom_point() +
  geom_path() +
  # plot thick black line for mean values
  geom_path(data = means_2026_summer, aes(x = depth_midpoint, y = mean_moisture), color = "black", size = 2) +
  coord_flip() +
  theme_cowplot() +
  scale_x_continuous(limits = c(-90,0), breaks = seq(-90, 0, by = 15)) +
  scale_y_continuous(limits = c(0,NA), position = "right") +
  labs(y = "% moisture", x = "Depth (cm)", title = "Mesa Grassland Cores, Summer 2026") +
  scale_color_ptol()
```

```
fig_2026_summer_moisture
```



```
ggsave(filename = here("figures/soil_cores/Summer_2026_Mesa_Grassland_moisture.pdf"),
        fig_2026_summer_moisture,
        width = 6,
        height = 4,
        units = "in",
        bg = "white")
```

Part 4: Plot Average Sand% and Average Clay % by Depth

```
# Texture ----

## Sand ----

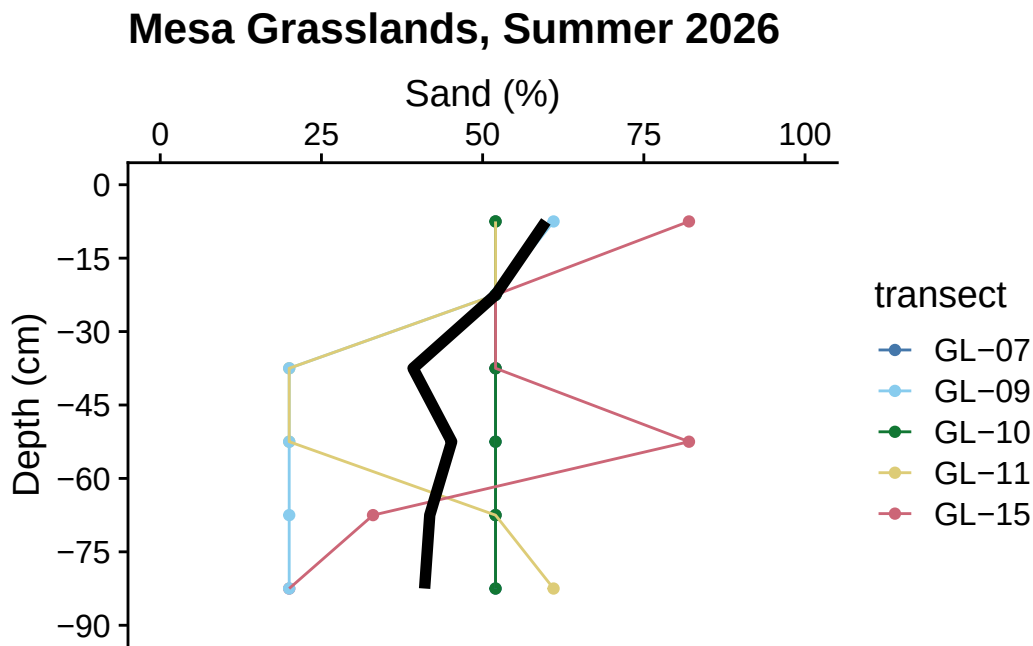
fig_2026_summer_sand <- ggplot(data = soil_data, aes(x = depth_midpoint, y = sand)) +
  geom_point(aes(color = transect, group = transect)) +
  #geom_jitter() +
  geom_path(aes(color = transect, group = transect)) +
  geom_path(data = means_2026_summer, aes(x = depth_midpoint, y = mean_sand), color = "black")
```

```

theme_cowplot() +
coord_flip() +
#ylab("Sample depth (cm)") +
#xlab("% Sand (estimated)") +
scale_x_continuous(limits = c(-90,0), breaks = seq(-90, 0, by =15)) +
scale_y_continuous(limits = c(0,100), position = "right") +
scale_color_ptol() +
labs(title = "Mesa Grasslands, Summer 2026", y = "Sand (%)", x = "Depth (cm)")

```

fig_2026_summer_sand



```

## Clay ----
fig_2026_summer_clay <- ggplot(data = soil_data, aes(x = depth_midpoint, y = clay)) +
  geom_point(aes(color = transect, group = transect)) +
  #geom_jitter() +
  geom_path(aes(color = transect, group = transect)) +
  geom_path(data = means_2026_summer, aes(x = depth_midpoint, y = mean_clay), color = "black") +
  theme_cowplot() +
  coord_flip() +
  #ylab("Sample depth (cm)") +
  #xlab("% Sand (estimated)") +
  scale_x_continuous(limits = c(-90,0), breaks = seq(-90, 0, by =15)) +

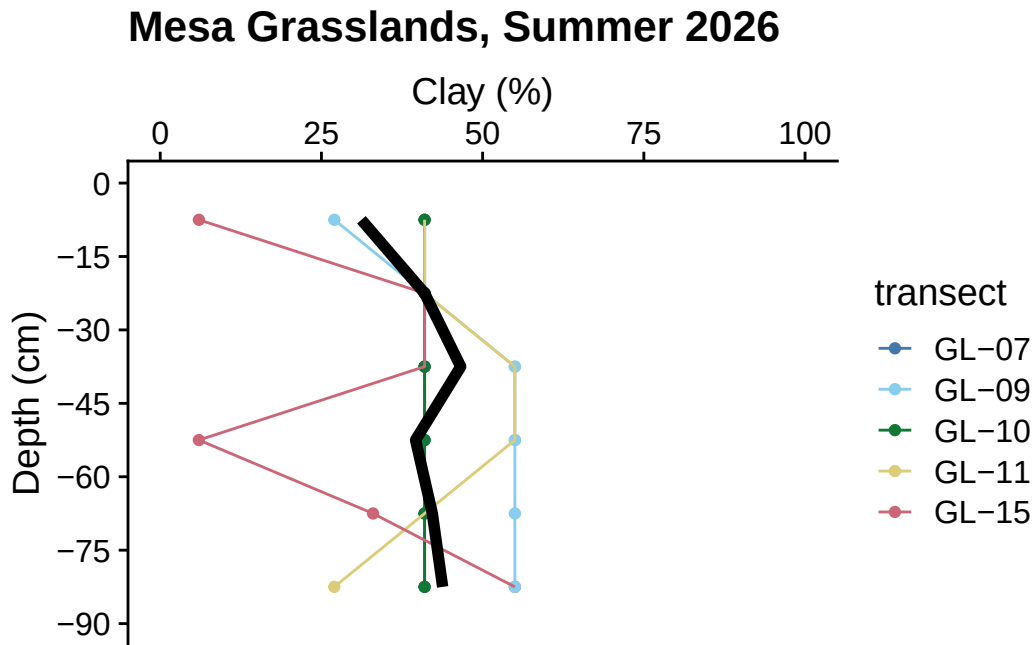
```

```

scale_y_continuous(limits = c(0,100), position = "right") +
scale_color_ptol() +
labs(title = "Mesa Grasslands, Summer 2026", y = "Clay (%)", x = "Depth (cm)")

fig_2026_summer_clay

```

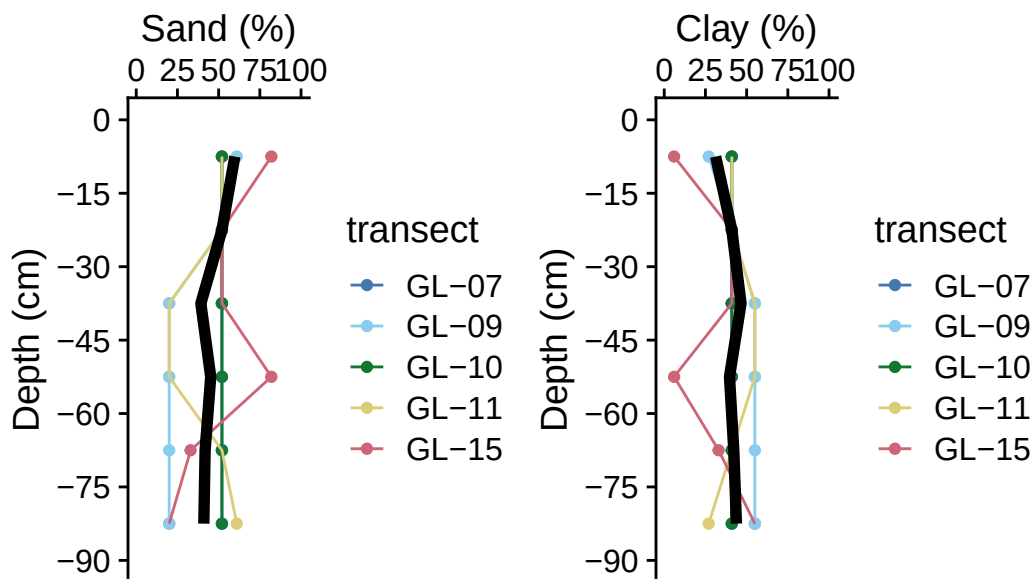


```

## combined plot ----
fig_slope_texture <- plot_grid(fig_2026_summer_sand, fig_2026_summer_clay,
                               nrow = 1)
fig_slope_texture

```

Mesa Grasslands, Summer 2026



Notes: * GL10 has pretty homogenous soil texture of sandy clay. Its soil moisture is relatively homogenous throughout the various depths. EC does not skew until last depth